

ABHISHEK JOSHI

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Google Scholar | Website

EDUCATION

PhD in Computer Science , Texas A&M University-Corpus Christi, Corpus Christi, Texas, 2025 - Present	GPA: 4.0
Master of Science in Computer Science , Texas A&M University-Corpus Christi, Corpus Christi, Texas, 2023 - 2025	GPA: 3.5

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, TypeScript

ML and AI: PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, OpenCV, RAG, Vector Databases, GNN

Robotics: ROS 2, DVS Event Cameras, LiDAR, Sensor Fusion, EKF Localization, LoRa Mesh Networking, Formal Verification (KeYmaera X)

Backend: Node.js, Express.js, Flask, Django, FastAPI, GraphQL

Frontend: React.js, Next.js, Vue.js, HTML5, CSS3, Tailwind CSS

Databases: PostgreSQL, MongoDB, Redis, Pinecone, ChromaDB, SQLite

DevOps and Cloud: AWS, GCP, Azure, Docker, Kubernetes, Terraform, CI/CD (Jenkins)

Version Control: Git, GitHub, GitLab

WORK EXPERIENCE

Doctoral Research Assistant, Texas A&M University Corpus Christi **Corpus Christi, Aug 2025 - Present**
Autonomy Research Institute, TAMUCC, ARI Fellow

- Engineered an event-triggered DVS, RGB, and LiDAR fusion pipeline on Raspberry Pi 5 embedded hardware, achieving greater than 85% defect detection accuracy in complete darkness at under 50ms end-to-end latency, reducing raw sensor throughput by 27,500x (220 MB/s to 8 kbps) to enable real-time transmission over LoRa bandwidth-constrained networks.
- Built SafeNav-RAG, a real-time retrieval-augmented decision framework for autonomous vehicles and UAVs in GPS-denied environments, reducing hallucination rates by 87% and improving response speed by 4.2x at sub-100ms latency across 10,000+ simulated safety-critical driving and aerial scenarios.
- Built a layered safety verification framework combining formal barrier certificate proofs (KeYmaera X theorem prover) for collision avoidance and geofencing, runtime safety wrappers constraining YOLO, PointNet, and GNN outputs, and statistical validation across 10,000 simulated mission scenarios, achieving zero safety violations with 99% statistical confidence.
- Collaborated with ARI faculty advisors and mentored five undergraduate engineering researchers for their capstone project.

Graduate Research Assistant, Texas A&M University Corpus Christi **Corpus Christi, Dec 2023 - Jul 2025**

- Optimized Transformer-based models for multi-modal stock forecasting, improving RMSE by 20% and inference time by 15% over LSTM baselines; results published at IEEE BigData 2024.
- Developed CryptoFusion, a temporal graph-aware model leveraging corporate relational networks, achieving a 7% gain in directional forecasting accuracy vs. LSTM; published at IEEE BigData 2025.
- Built and deployed a real-time portfolio rebalancing engine on university research infrastructure supporting 50+ trading strategies, reducing computation costs by 40%.

Senior Software Engineer, Kellton **India, Feb 2022 - Aug 2023**

- Architected microservices infrastructure processing 10,000+ requests per second, reducing API response time by 35% through Redis caching and asynchronous processing pipelines, directly improving SLA compliance for 3 enterprise clients and enabling 4x faster feature delivery to production; owned technical design documentation and led architecture reviews with cross-functional teams.
- Designed and owned an end-to-end CI/CD pipeline achieving 98% deployment success rate, compressing release cycles from 2 weeks to 3 days using Jenkins and Docker with automated regression test suites.
- Led large-scale codebase refactoring applying SOLID principles across 150,000+ lines of legacy code, reducing technical debt by 40%, cutting critical bug rate by 30%, and elevating test coverage from 45% to 85%, reducing QA cycle time by 2 weeks per release; conducted weekly code reviews with junior engineers to enforce standards across the team.

Senior Software Engineer, Star World Digital **India, Dec 2020 - Feb 2022**

- Led a team of 8 engineers in architecting a multi-tenant SaaS platform from the ground up, improving system throughput by 33% via PostgreSQL query optimization and connection pooling, scaling to support 3x user growth without infrastructure changes; owned end-to-end technical design docs and drove cross-functional alignment with product and QA teams.
- Engineered a real-time customer analytics dashboard processing 1M+ daily events with sub-second query response, enabling data-informed product decisions that improved user retention by 25% and reduced churn by 18% within 2 quarters of deployment.
- Achieved and maintained 95% uptime SLA across distributed services by implementing proactive monitoring with AWS CloudWatch and automated failover, reducing mean time to recovery (MTTR) by 60% vs. prior manual incident response.

PROJECTS

Business Build AI **Corpus Christi, TX, Jan 2025 - May 2025**

(LLMOps | OpenAI GPT-4 | Stable Diffusion | Node.js | TypeScript | AWS S3 | Cloudflare | React.js)

- Developed an AI-powered platform automating brand asset generation, reducing design time by 40% and manual content production by 50%.
 - Implemented scalable architecture using AWS S3 and Cloudflare CDN, achieving 30% faster page load times and 99.9% uptime.
- [Link](#)

(TensorFlow | NLP | OpenCV | React.js | Node.js | MongoDB)

- Built an AI platform converting text to video and image content, increasing content creation efficiency by 50% for 100+ users.
- Developed a custom NLP pipeline for text analysis and scene generation, achieving 85% accuracy in content relevance. [Link](#)

PUBLICATIONS

- Joshi, A., Koda, J. K., and Phadke, A. "Sequence-Preserving Dual-FoV Defense for Traffic Sign and Light Recognition in Autonomous Vehicles." MDPI *Sensors*, 2026. [Link](#)
- Joshi, A., Verma, N. K., and Hadimlioglu, A. "SafeNav-RAG: A Latency-Aware Retrieval-Augmented Generation Framework for Autonomous Vehicle Decision-Making." *IEEE UPCON*, 2025
- Joshi, A., Hadimlioglu, A., and Verma, N. K. "CryptoFusion: A Temporal Graph-Aware Transformer Framework for Robust Cryptocurrency Forecasting and Adaptive Portfolio Optimization." *IEEE BigData*, 2025. [Link](#)
- Joshi, A., Koda, J. K., Hadimlioglu, A., and Kaur, G. "Real-Time Adaptive Multi-Modal Stock Prediction with Temporal Graph Attention and Dynamic Interaction Networks." *IEEE CAI*, 2025. [Link](#)
- Joshi, A., Koda, J. K., and Hadimlioglu, A. "A Multi-Modal Transformer Architecture Combining Sentiment Dynamics, Temporal Market Data, and Macroeconomic Indicators for Sturdy Stock Return Forecasting." *IEEE BigData*, 2024. [Link](#)

AWARDS and ACHIEVEMENTS

- [National Science Foundation](#) Student Award for [FLAIRS](#) 2024 - one of 5 nationwide, for distinguished achievements in machine learning research.
- [ARI](#) TAMUCC Fellowship 2025 - Competitively awarded to fund T-AMASS doctoral research in trusted autonomous swarm systems for GPS-denied multi-domain environments.
- College of Engineering Graduate Scholarship, TAMUCC 2025 - Awarded for distinguished academic achievement and research excellence.
- Session Chair, [IEEE CAI 2025](#) - Chaired the research track for vertical business intelligence.
- Peer Reviewer, [DMI 2025](#) - Assessed three AI papers on LLM and RAG chatbots, chain-of-thought reasoning, and epidemic-informed PINNs.
- Grand Finalist, [Smart India Hackathon](#) 2019 - Selected from among 200,000+ students. Built an EV range monitoring solution for Mahindra Electric.
- [IEEE XTREME](#) - All India rank 105 in 2018 and rank 109 in 2019.

INTERNSHIP EXPERIENCE

Software Engineer Intern, AIMAKERY

London, Sep 2019 - Dec 2020

- Developed core modules for decentralized cryptocurrency exchange, handling 50,000+ daily transactions with 99.9% accuracy.
- Built a custom CMS portal using React and Node.js, reducing content management time by 40% for the marketing team.

Software Engineer Intern, MOLOG MEDIA

India, Sep 2018 - Sep 2019

- Improved website performance by 60% through code optimization, lazy loading, and CDN implementation.
- Built a visitor management system tracking 10,000+ monthly visitors, reducing check-in time by 50% through QR code integration.